CO1 - Assessment Tool 1

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Course code: DSA0503
Course Name: Query Processing for Data Science

Aim

To read data from an XML file (`library.xml`) and display:

- Total number of books
- Books published after 2020
- Books costing more than ₹500

EXPLANATION

- `import xml.etree.ElementTree as ET` – Imports the XML parsing module.
- `ET.parse("library.xml")` – Loads the XML file.
- `getroot()` – Retrieves the root element of the XML.
- `findall("book")` – Finds all book elements.
- `len()` – Counts the total number of books.
- `for book in root.findall("book")` – Iterates through each book.
- `book.find("year").text` – Reads the publication year.
- `book.find("price").text` – Reads the book price.
- `if year > 2020` – Counts books published after 2020.
- `if price > 500` – Counts books costing more than ₹500.
- `print()` – Displays the total books, books after 2020, and books costing more than ₹500.

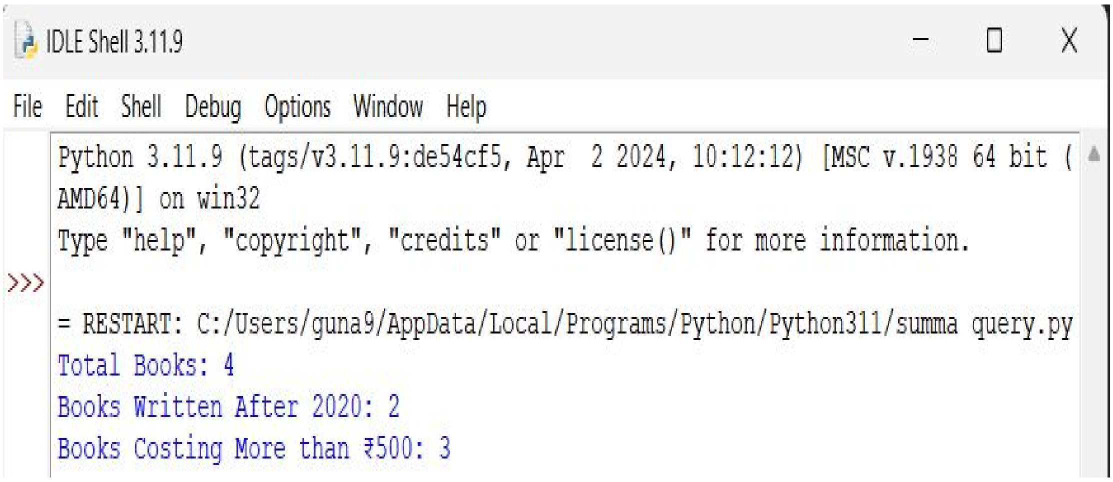
ALGORITHM

- Import the `xml.etree.ElementTree` module.
- Load and parse the `library.xml` file.
- Get the root element of the XML document.
- Count the total number of `<book>` elements.
- Traverse each book element.
- Check if the publication year is greater than 2020 and count those books.
- Check if the price is greater than ₹500 and count those books.
- Display the required results.

CODE :

```
import xml.etree.ElementTree as ET

xml_data = """
<library>
   <book>
      <title>Python Programming</title>
      <author>John Smith</author>
      <year>2019</year>
      <price>450</price>
   </book>
   <book>
      <title>Data Science Essentials</title>
      <author>David Miller</author>
      <year>2021</year>
      <price>650</price>
   </book>
   <book>
      <title>Artificial Intelligence</title>
      <author>James Brown</author>
      <year>2023</year>
      <price>750</price>
   </book>
   <book>
      <title>Machine Learning Basics</title>
      <author>Emily Davis</author>
      <year>2020</year>
      <price>550</price>
   </book>
</library>
"""

root = ET.fromstring(xml_data)

total_books = len(root.findall("book"))
books_after_2020 = 0
books_costing_more_500 = 0

for book in root.findall("book"):
   year = int(book.find("year").text)
   price = float(book.find("price").text)

   if year > 2020:
      books_after_2020 += 1

   if price > 500:
      books_costing_more_500 += 1

print("Total Books:", total_books)
print("Books Written After 2020:", books_after_2020)
print("Books Costing More than ₹500:", books_costing_more_500
```

OUTPUT


IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/guna9/AppData/Local/Programs/Python/Python311/summa query.py
Total Books: 4
Books Written After 2020: 2
Books Costing More than ₹500: 3